

ChengYen Kuo

+1 206-251-0324 | ckuo5688@uw.edu | linkedin.com/in/chengyen-kuo | Seattle, WA

Graduate student in Electrical Engineering at the University of Washington with hands-on experience in semiconductor fabrication process (ISO Class 5) and advanced TCAD modeling. Skilled in core processes (Lithography, PECVD, RIE), device characterization (SEM, AFM, I-V), and root-cause analysis. Eager to contribute technical expertise to process optimization and yield enhancement in semiconductor manufacturing.

EDUCATION

University of Washington

Seattle, WA

M.S. in Electrical & Computer Engineering | GPA: 3.8/4.0

Expected Jun. 2026

Courses: MEMS, Solid-State Electronics, Optoelectronic Materials, Microfabrication, Integrated Circuit Technology, Silicon Photonics, Database Systems

Chung Yuan Christian University (CYCU)

Taoyuan, Taiwan

B.S. in Electronic Engineering & Computer Science | GPA: 3.7/4.0

Sep. 2019 – Jun. 2023

Courses: Nano-electronics, Semiconductor Devices/Physics, Electronics, Circuit Theory, Java, Digital Systems, Logic Design, Signal Integrity for High-Speed Digital Circuits

TECHNICAL SKILLS

Fabrication & Characterization: CVD, PECVD, PVD, RIE, Dry/Wet Etching, Photolithography, SEM, AFM, profilometry

Simulation & Modeling: Ansys Lumerical, Ansys Maxwell, Sentaurus TCAD, SolidWorks (CSWA certification)

Programming: Java, C/C++, Python, MATLAB, SQL

PROJECT EXPERIENCE

Washington Nanofabrication Facility (WNF)

Seattle, WA

Microfabrication Project

Jan. 2025 – Apr. 2025

- Fabricated microelectronic devices by integrating **PECVD** (465 nm SiO_2), advanced photolithography, and **Reactive Ion Etching (RIE)** with CF_4/CHF_3 , achieving precise Critical Dimension (CD) control for high-resolution features.
- Engineered robust metallization stacks utilizing **E-beam Evaporation** (Cr/Al) and **Sputtering** (Al/Cu), optimizing step coverage and electromigration resistance for reliable interconnects.
- Validated process quality by achieving $\pm 0.4\%$ oxide uniformity (verified via Filmetrics) and performed Electrical Failure Analysis (EFA) using diode IV/CV sweeps to correlate process defects with device performance.

EXPERIENCE

Nano Optoelectronic Integrated System Lab (UW)

Seattle, WA

Graduate Researcher

Sep. 2025 – Present

- Performed electro-optical device simulations of an InGaAsP/InP Micro-LED integrated with a heterojunction phototransistor using **Lumerical**, modeling carrier transport via drift-diffusion and quantum-well recombination in MQW structures to extract spontaneous emission characteristics.
- Developed Python post-processing pipelines to evaluate Internal Quantum Efficiency (IQE) and Light Extraction Efficiency (LEE), combining emission data with optical simulations to estimate total emitted optical power for a $300\text{ }\mu\text{m} \times 300\text{ }\mu\text{m}$ device footprint.
- Investigated the impact of surface recombination velocity on device efficiency, modeling sidewall defect effects associated with mesa formation and plasma damage from Reactive Ion Etching to analyze efficiency degradation mechanisms in Micro-LED structures.

Nano Optoelectronic Materials Lab (CYCU)

Taoyuan, Taiwan

Undergraduate Researcher

Jun. 2021 – Oct. 2022

- Synthesized diamond films on silicon via **Hot-Filament CVD**, optimizing gas ratios (CH_4/H_2), pressure, and temperature for DUV applications.
- Performed Physical Failure Analysis (PFA) using Raman Spectroscopy (identifying the 1332 cm^{-1} peak), **SEM** and **AFM** to characterize surface morphology defects, correlating roughness data with deposition parameters to optimize film quality.
- Developed MATLAB algorithms to mine spectroscopic data, identifying trend lines that guided the optimization of gas ratios for enhanced thermal stability.

Military Service

Tainan, Taiwan

Private

Dec. 2023 – Apr. 2024